

# Chapter 12

## Potential Energy

# The Energy in the World

Thermal Energy

Electrical Energy

Atomic Energy

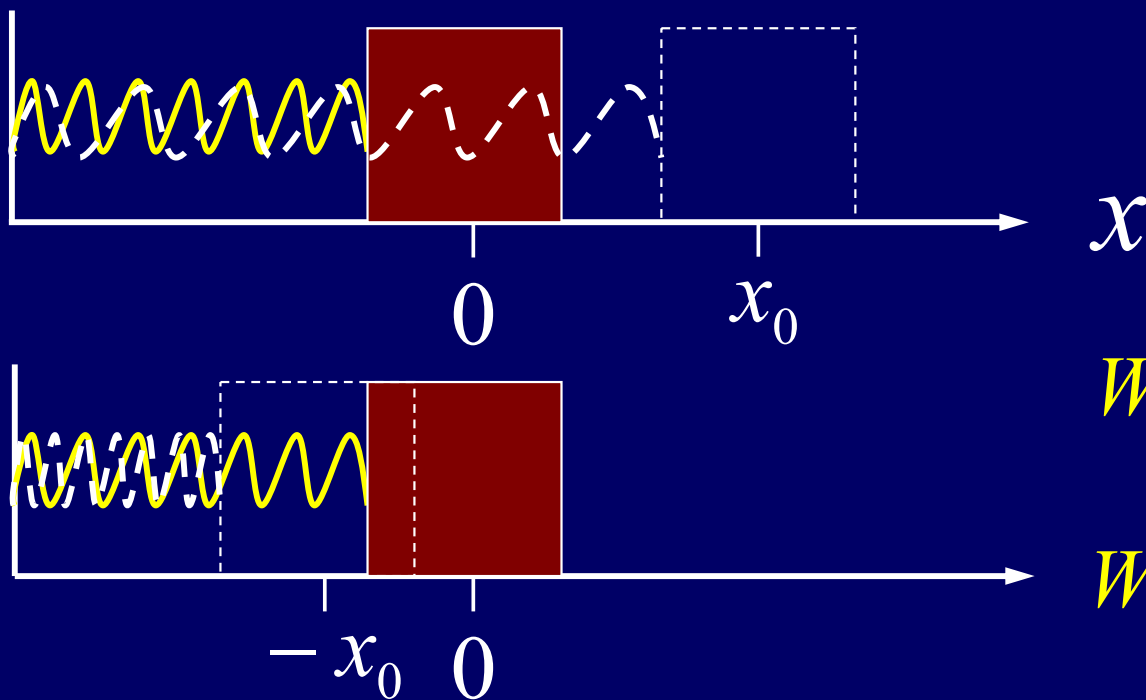
Chemical Energy  $\frac{1}{2}mv^2$   $\frac{1}{2}I\omega^2$   $\frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$

Mechanical Energy { Kinetic Energy  
Potential Energy

Conservative forces

# Conservative forces

$$\oint \vec{F} \cdot d\vec{s} = 0$$



$$W = \int_0^{x_0} -kx dx = -\frac{1}{2} kx_0^2$$

$$W' = \int_{x_0}^0 -kx dx = \frac{1}{2} kx_0^2$$

$$W + W' = 0$$

$$W = \int_0^{-x_0} -kx dx = -\frac{1}{2} kx_0^2$$

$$W' = \int_{-x_0}^0 -kx dx = \frac{1}{2} kx_0^2$$

$$W + W' = 0$$

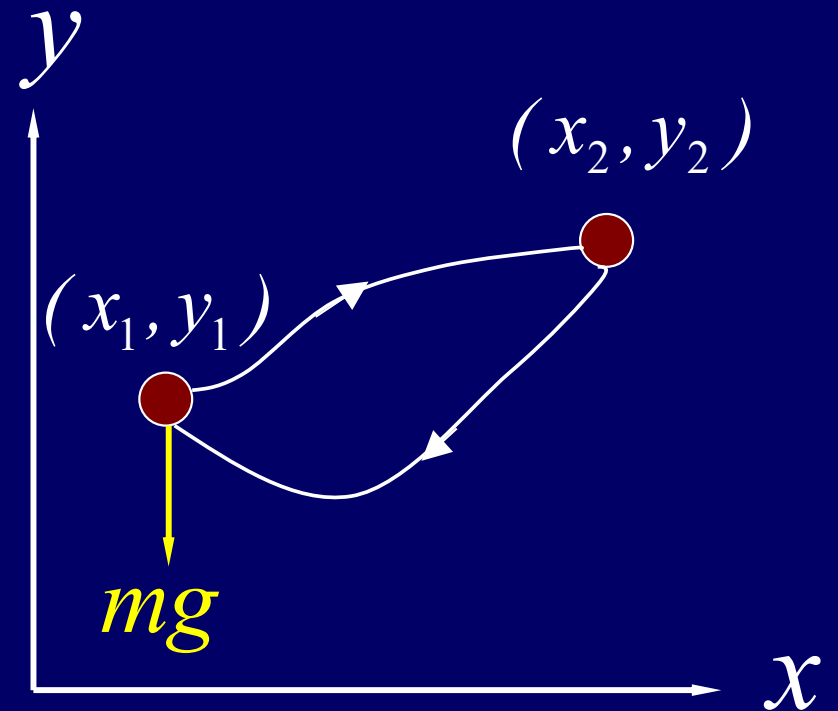
zero total work during any round trip

$$W = \int_0^{z_0} -mgdz = -mgz_0$$

$$W' = \int_{z_0}^0 -mgdz = mgz_0$$

$$W + W' = 0$$

zero total work during  
any round trip



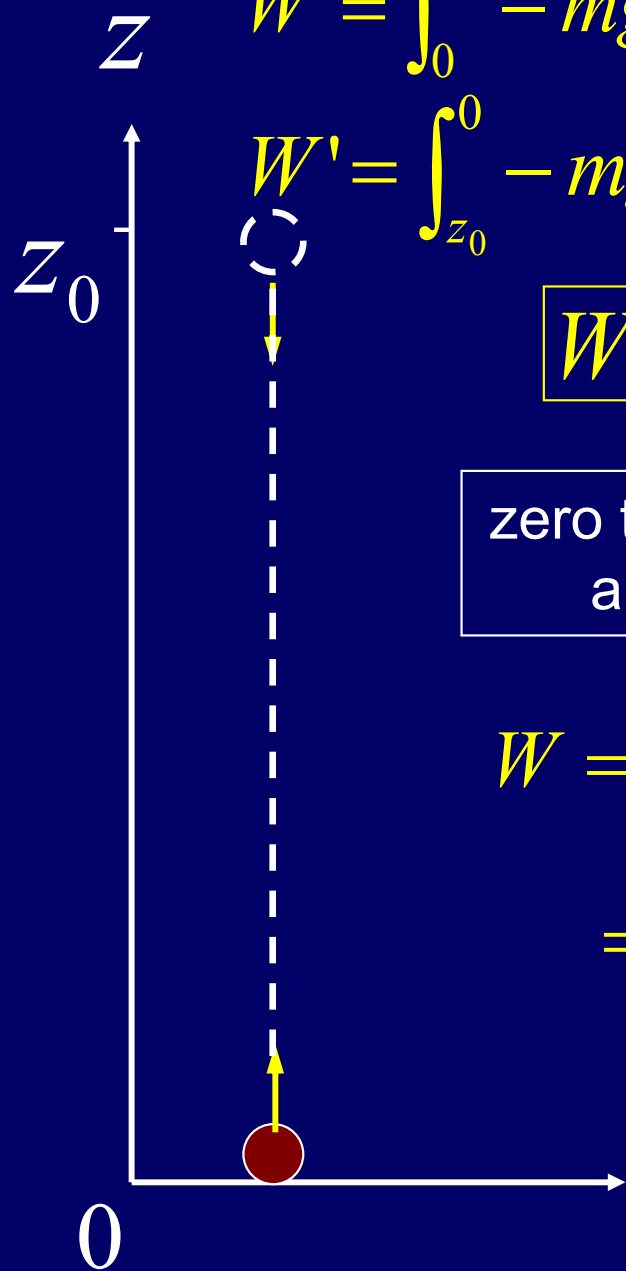
$$W = \int_i^f (F_x \bar{i} + F_y \bar{j}) \cdot (dx \bar{i} + dy \bar{j}) = \int_i^f F_x dx + F_y dy$$

$$= \int_{y_1}^{y_2} -mgdy = mg(y_1 - y_2)$$

$$W' = \int_{y_2}^{y_1} -mgdy = mg(y_2 - y_1)$$

$$W + W' = 0$$

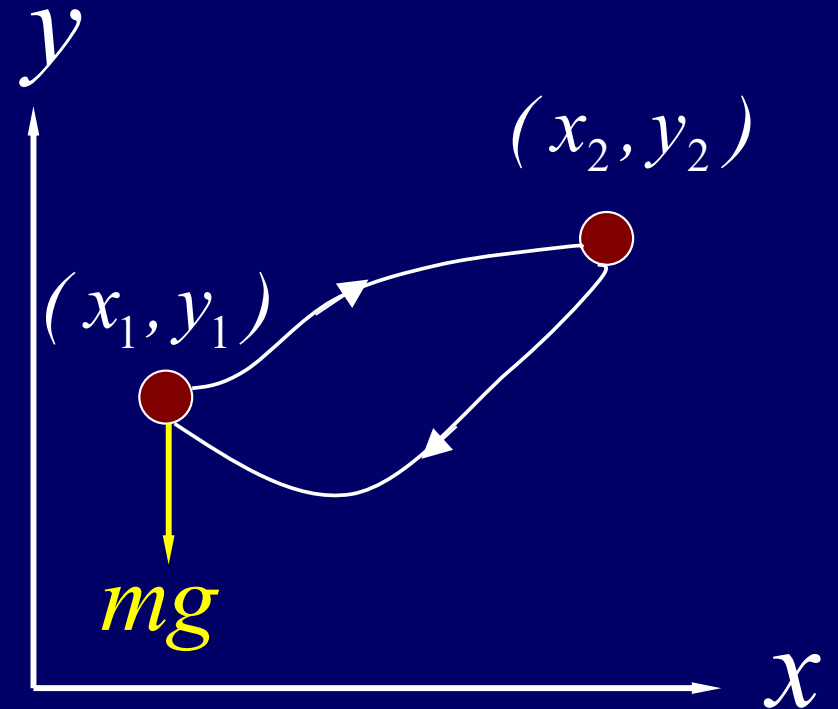
zero total work during  
any round trip



If a body moves under a force that does **no total work during any round trip**.

$$\oint \vec{F} \cdot d\vec{s} = 0$$

conservative force



The spring force, gravity and universal gravitation are three examples of **conservative force**.

$$(F_x \vec{i} + F_y \vec{j}) \cdot (dx \vec{i} + dy \vec{j}) = \int_i^f F_x dx + F_y dy$$

$$-mgdy = mg(y_1 - y_2)$$

$$W' = \int_{y_2}^{y_1} -mgdy = mg(y_2 - y_1)$$

$$W + W' = 0$$

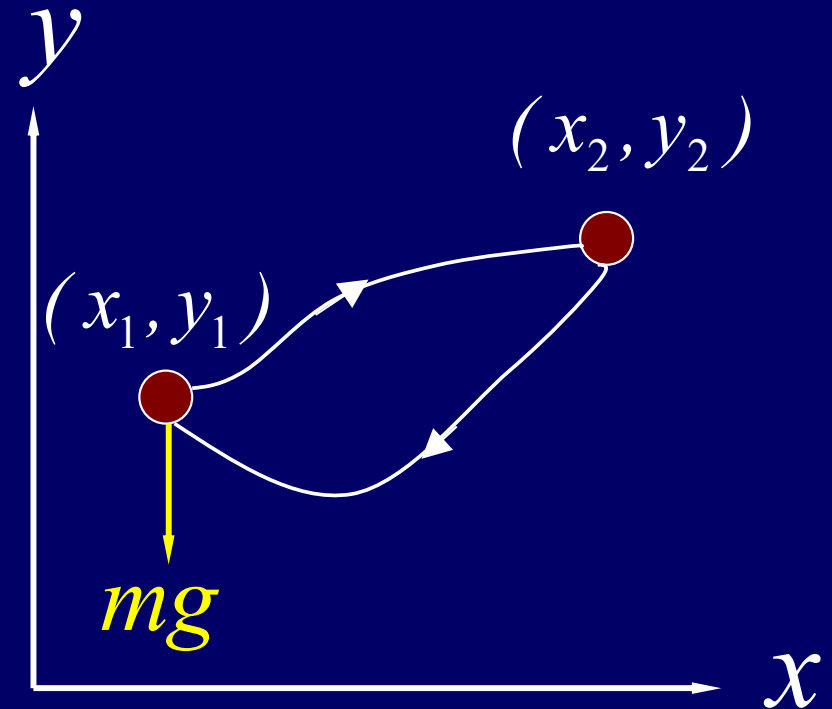
zero total work during any round trip

If a body moves under a force that does **no total work during any round trip**.

$$\oint \vec{F} \cdot d\vec{s} = 0$$

conservative force

The spring force, gravity and universal gravitation are three examples of **conservative force**.



$$\int_{x_1 y_1}^{x_2 y_2} \vec{F} \cdot d\vec{s} + \int_{x_2 y_2}^{x_1 y_1} \vec{F} \cdot d\vec{s} = 0$$

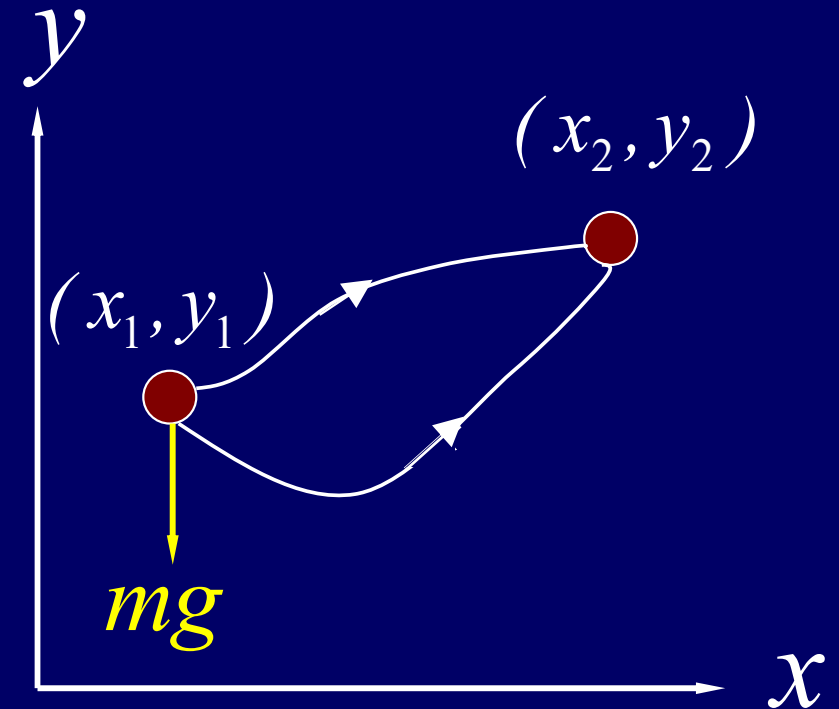
$$\int_{x_1 y_1}^{x_2 y_2} \vec{F} \cdot d\vec{s} - \int_{x_1 y_2}^{x_2 y_2} \vec{F} \cdot d\vec{s} = 0$$

If a body moves under a force that does no total work during any round trip.

$$\oint \vec{F} \cdot d\vec{s} = 0$$

conservative force

If the work done by a force is independent of the path. This force is a conservative force



$$\int_{x_1 y_1}^{x_2 y_2} \vec{F} \cdot d\vec{s} + \int_{x_2 y_2}^{x_1 y_1} \vec{F} \cdot d\vec{s} = 0$$

$$\int_{x_1 y_1}^{x_2 y_2} \vec{F} \cdot d\vec{s} = \int_{x_1 y_2}^{x_2 y_2} \vec{F} \cdot d\vec{s}$$

The spring energy,  
gravity and universal  
gravitation energy.

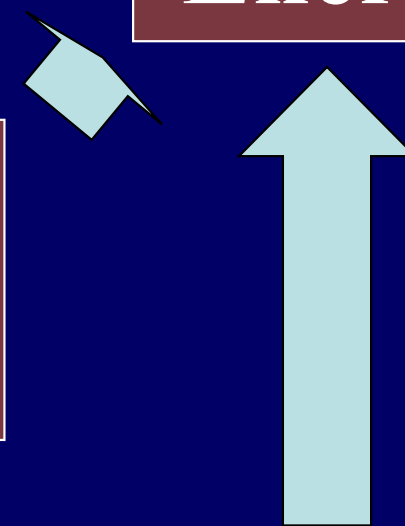
Potential  
Energy

Energy

Work  
done by conservative force

independent of  
the path

dependent on  
the position





Because the work done by the conservative force is independent of the path taken between the two point, we can introduce a new concept,

## **potential energy**

represented by the symbol  $U$ .

We define the change in potential energy associated with a conservative force as:

$$- \Delta U = U_i - U_f = W$$

$$\Delta U = U_i - U_f = - W$$

For one dimensional motion, a conservative force  $F(x)$  acts on a particle that moves from the initial coordinate  $x_0$  to the final coordinate  $x$ .

$$\Delta U = U(x) - U(x_0) = -\int_{x_0}^x F(x) dx = \int_x^{x_0} F(x) dx$$

$$U(x) = \int_x^{x_0} F(x) dx + U(x_0)$$

Here  $U(x_0)$  is the potential energy of an arbitrary **reference point**. It can be free to choose any convenient value because it can not change the difference of the potential energy for two locations.

$$U_1(x_1) = \int_{x_1}^{x_0} F(x) dx + U(x_0) \quad U_2(x_2) = \int_{x_2}^{x_0} F(x) dx + U(x_0)$$

$$U_2(x_2) - U_1(x_1) = \int_{x_2}^{x_1} F(x) dx$$

The spring force :  $F(x) = -kx$

$$U(x) = \int_x^{x_0} -kx dx + U(x_0) = \int_x^0 -kx dx + U(x_0) = \frac{1}{2}kx^2$$

If we choose the reference position  $x_0$  of the block in the spring being its relaxed state ( $x_0 = 0$ ), then  $U(x_0) = 0$ .

The force of gravity:  $F(z) = -mg$

$$U(z) = \int_z^{z_0} -mg dz + U(z_0) = \int_z^0 -mg dz + U(z_0) = mgz$$

If we choose the reference point  $y_0$  at the surface of the earth and  $z_0 = 0$ , then we have  $U(z_0) = 0$ .

The relation between **force** and **potential energy**:

$$-\Delta U = \int_{x_0}^x F(x)dx \quad -dU = F(x)dx \quad F(x) = -\frac{dU(x)}{dx}$$

conservative force



potential energy

Force of gravity

$$F(y) = -\frac{dU(y)}{dy} = -\frac{d(mgy)}{dy} = -mg$$

Spring Force

$$F(x) = -\frac{dU(x)}{dx} = -\frac{d(\frac{kx^2}{2})}{dx} = -kx$$

# Three-dimensional conservative system

A particle moves along a path from  $i$  to  $f$ . A conservative force  $F$  acts on the particle.

$$\Delta U = U(x_f, y_f, z_f) - U(x_i, y_i, z_i) = -\int_i^f \vec{F} \cdot d\vec{r} \quad F_x = -\frac{dU}{dx}$$

$$\int dU = -\int (F_x dx + F_y dy + F_z dz) \quad F_y = -\frac{dU}{dy}$$

$$\vec{F}(x, y, z) = -\frac{\partial U}{\partial x} \vec{i} - \frac{\partial U}{\partial y} \vec{j} - \frac{\partial U}{\partial z} \vec{k} = -\nabla U \quad F_z = -\frac{dU}{dz}$$

$F$  is the negative gradient of the potential energy  $U(x, y, z)$ .

conservative force



potential energy

Example:

$$U = 2x - y^3 + 2z^2$$

$$\vec{F} = -\frac{\partial U}{\partial x} \vec{i} - \frac{\partial U}{\partial y} \vec{j} - \frac{\partial U}{\partial z} \vec{k}$$

$$= -2\vec{i} + 3y^2\vec{j} - 4z\vec{k}$$

# Conservation of mechanical energy

Consider an isolated system: The internal forces are conservational forces

$$W = -\Delta U \quad W = \Delta K$$

$$\Delta K = -\Delta U \quad \Delta(K+U) = 0$$

$$K+U=E=\text{constant}$$

$E$  is called mechanical energy of the conservative system.

In moving from  $x_0$  to  $x$ , the velocity of a particle it is acted on a conservative force will change from  $v_0$  to  $v$ .

$$\Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 \quad \Delta U = U(x) - U_0(x_0)$$

$$\Delta(K + U) = 0$$

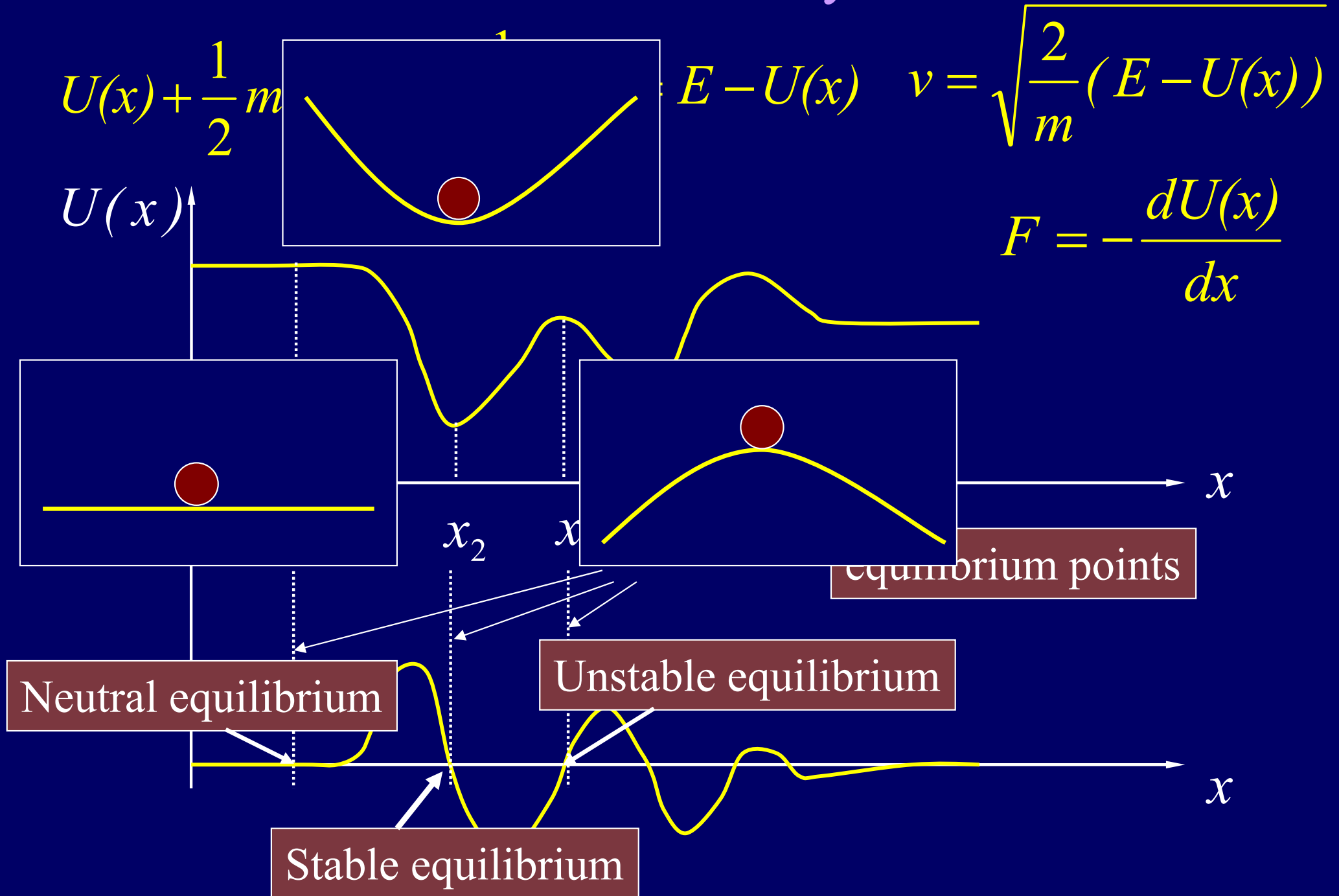
$$\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 + U(x) - U(x_0) = 0$$

$$\frac{1}{2}mv^2 + U(x) = \frac{1}{2}mv_0^2 + U(x_0) = E$$

**Conservation of mechanical energy**

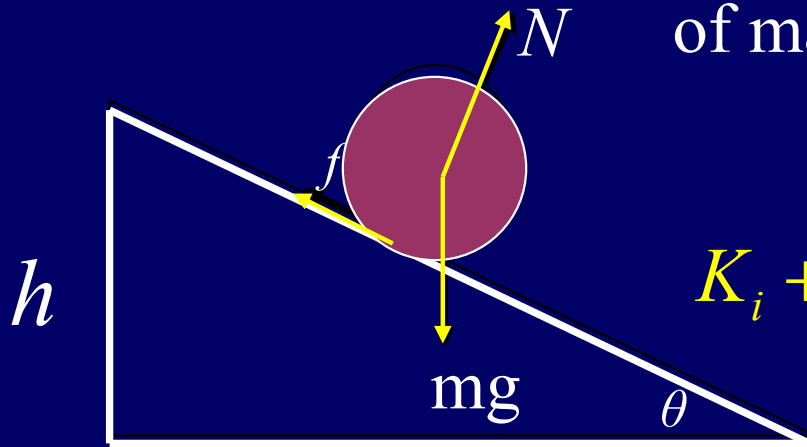


# One-dimensional conservation systems:



Example:

Find  $v$  of the rolling cylinder when the center of mass move down from height  $h$  at the rest.



$$K_i + U_i = mgh \quad K_f + U_f = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} M v_{cm}^2$$

$$\frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} M v_{cm}^2 = mgh$$

$$mg \sin \theta - f = m a_{cm}$$

$$fR = \frac{1}{2} m R^2 \alpha \quad a_{cm} = R \alpha$$

$$a_{cm} = \frac{2g \sin \theta}{3}$$

$$v^2 = 2 a_{cm} L = \frac{4gh}{3}$$

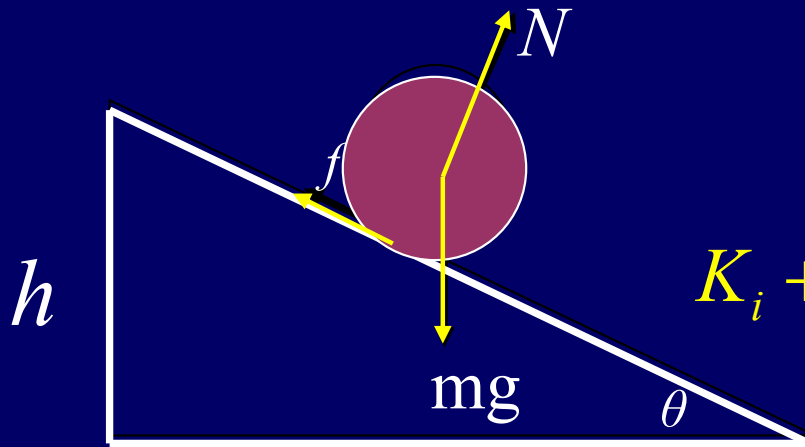
$$\frac{1}{4} m v^2 + \frac{1}{2} m v^2 = mgh$$

$$\frac{3}{4} m v^2 = mgh$$

$$v = \sqrt{\frac{4gh}{3}}$$

Example:

Find  $v$  of the rolling ball  
when the center of mass move  
down from height  $h$  at the rest.



$$K_i + U_i = mgh$$

$$K_f + U_f = \frac{1}{2} I \omega^2$$

$$mgh = \frac{1}{2} \left( \frac{3}{2} m R^2 \right) \omega^2 = \frac{3}{4} m v^2$$

$$mg \sin \theta - f = m a_{cm}$$

$$f R = \frac{1}{2} m R^2 \alpha$$

$$a_{cm} = \frac{2g \sin \theta}{3}$$

$$v^2 = 2 a_{cm} s = \frac{4gh}{3}$$

$$v = \sqrt{gh}$$

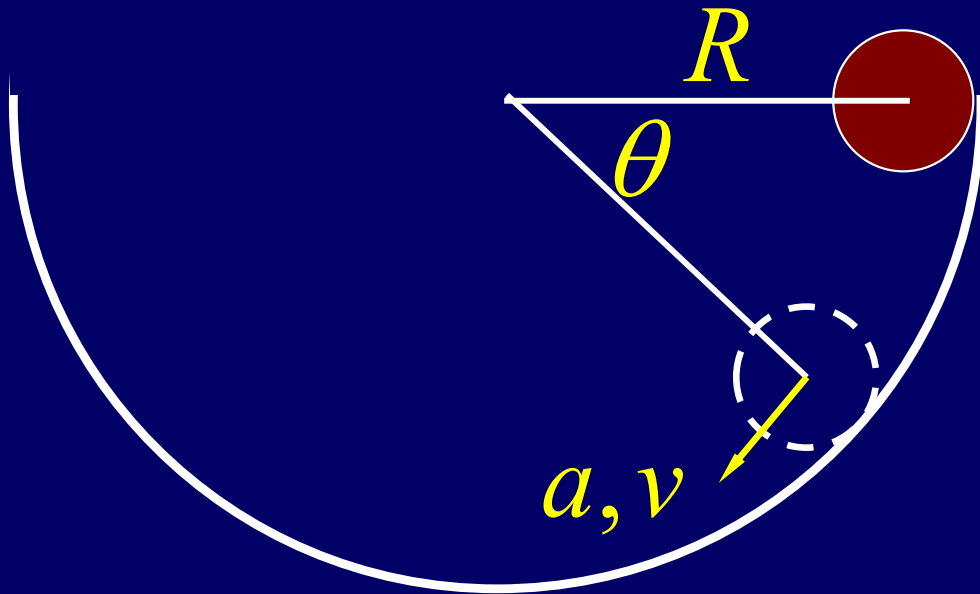
$$v = \sqrt{\frac{4gh}{3}}$$

$$v = \sqrt{\frac{10gh}{7}}$$

$$mgh = \frac{1}{2} (2mR^2) \omega^2 = mv^2$$

$$mgh = \frac{1}{2} \left( \frac{7}{5} m R^2 \right) \omega^2 = \frac{7}{10} m v^2$$

## Example

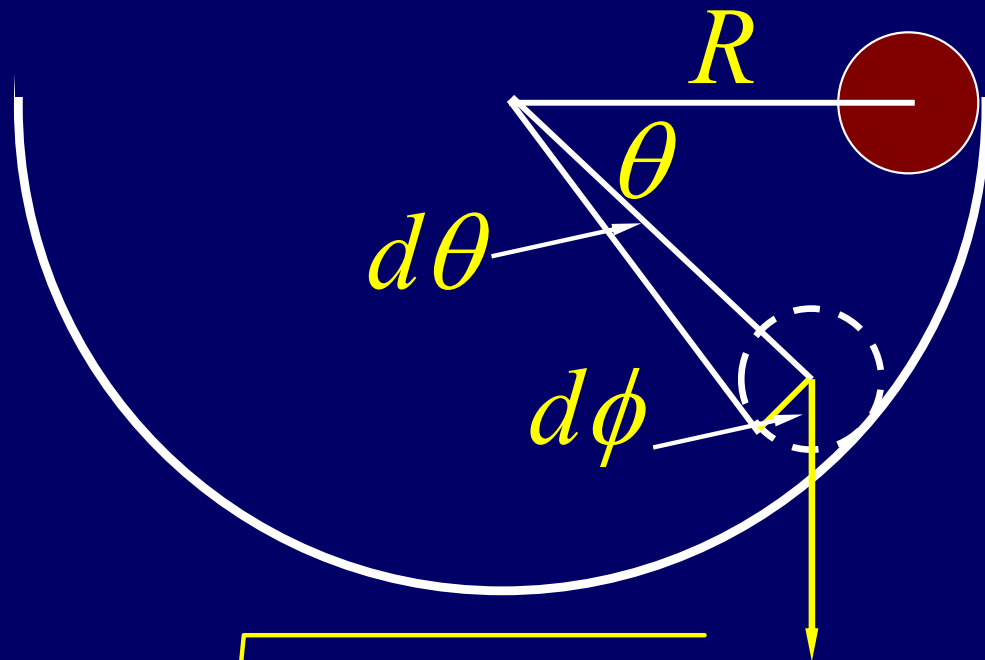


A solid ball with mass  $m$  and radius  $r$

$$r \ll R$$

**Rolling without slipping**

Example



$$\omega = \sqrt{\frac{10gR \sin \theta}{7r^2}} \quad mg$$

$$v = r\omega = \sqrt{\frac{10gR \sin \theta}{7}}$$

$$mgr \cos \theta = \frac{7}{5} mr^2 \alpha$$

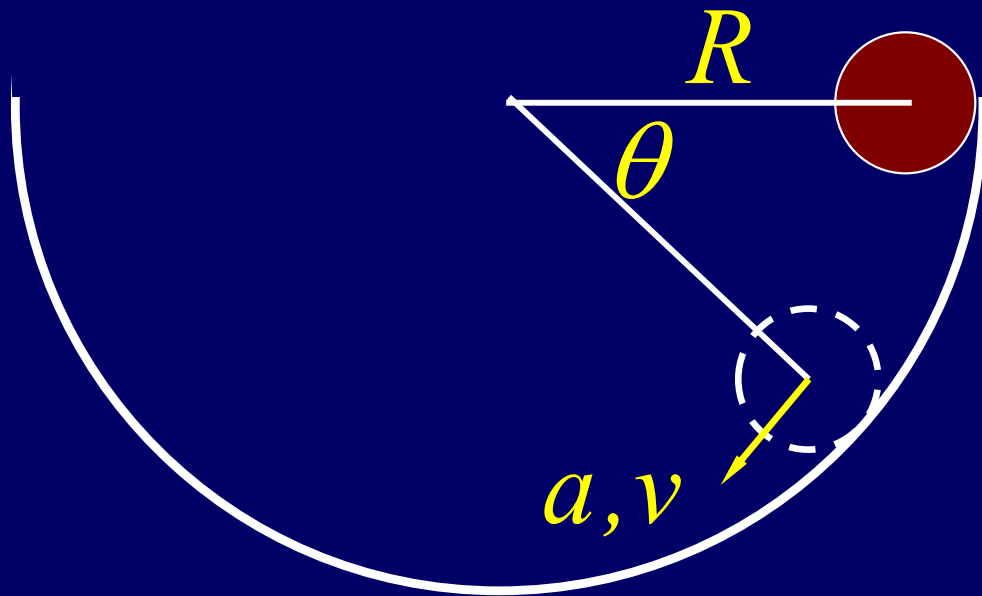
$$\alpha = \frac{5g \cos \theta}{7r}$$

$$a_T = \alpha r = \frac{5g \cos \theta}{7}$$

$$a_R = \frac{v^2}{R} = \frac{10g \sin \theta}{7}$$

$$\int_0^\omega \omega d\omega = \int_0^\theta \frac{5gR \cos \theta d\theta}{7r^2}$$

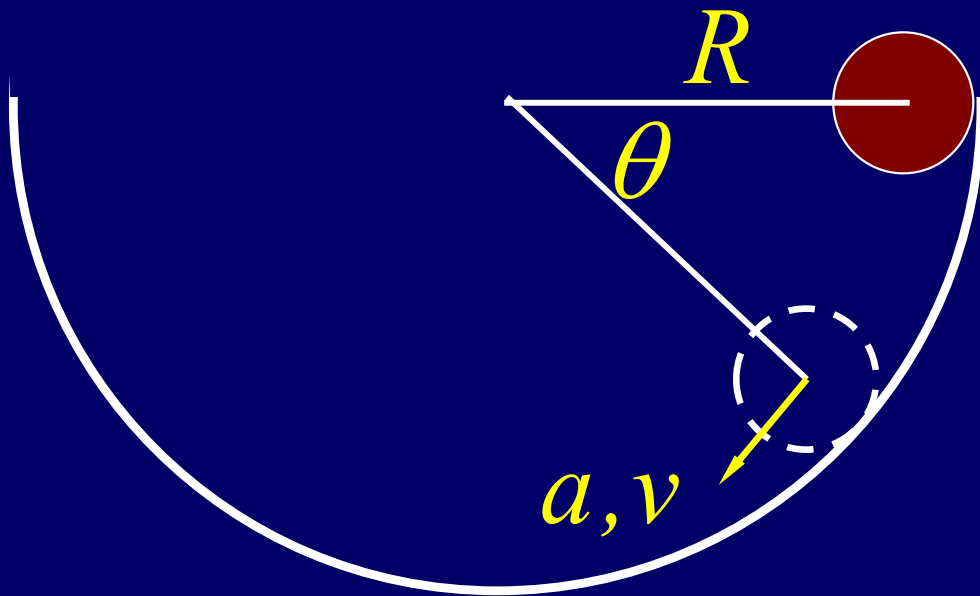
Example



$$\frac{7}{10}mv^2 = mgR \sin \theta$$

$$v = \sqrt{\frac{10gR \sin \theta}{7}}$$

Example



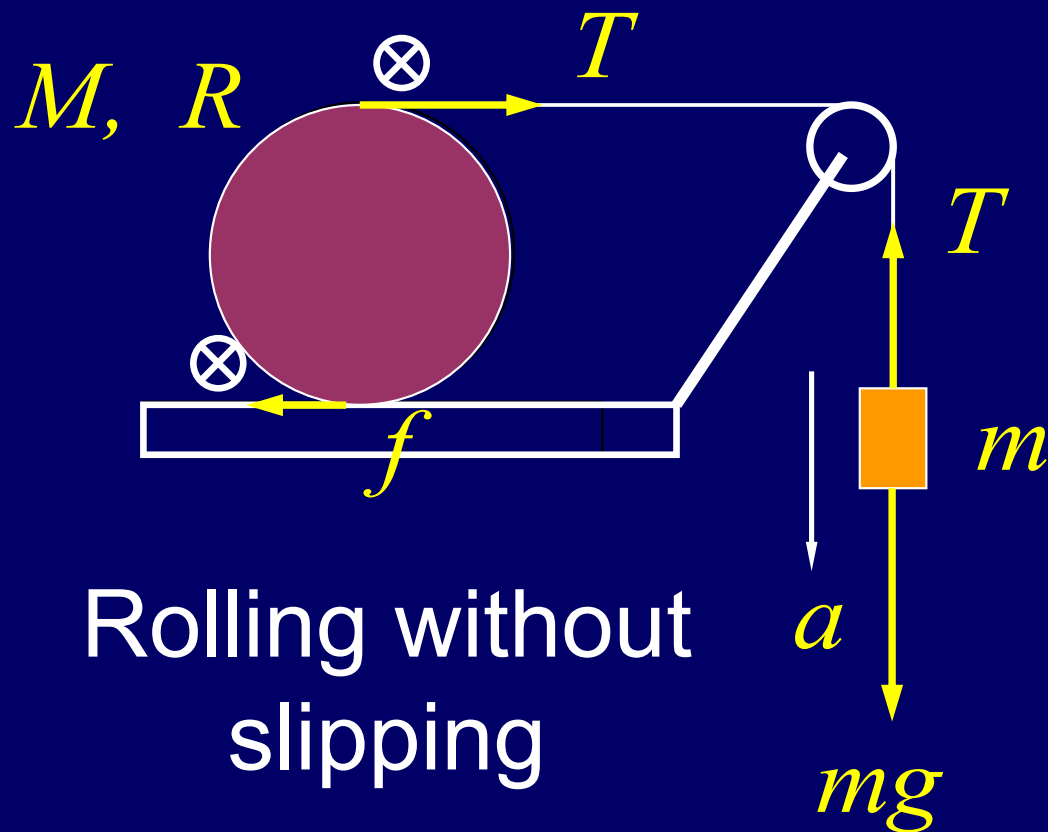
$$\frac{1}{2} \left( \frac{7}{5} m r^2 \right) \omega^2 = m g R \sin \theta$$

$$v = \sqrt{\frac{10 g R \sin \theta}{7}}$$

$$\alpha_R = \frac{v^2}{R} = \frac{10 g \sin \theta}{7}$$

$$\begin{aligned} a_T &= \frac{dv}{dt} = \sqrt{\frac{10 g R}{7}} \frac{\cos \theta}{2 \sqrt{\sin \theta}} \frac{v}{R} \\ &= \frac{5 g \cos \theta}{7} \end{aligned}$$

## Example



$a = ?$

$$mg = ma + \frac{3}{8}Ma$$

$$T - f = Ma_{cm}$$

$$T = \frac{3}{8}Ma$$

$$\alpha R = a_{cm} \quad a = a_{cm} ?$$

$$a = a_{cm} + \alpha R = 2a_{cm}$$

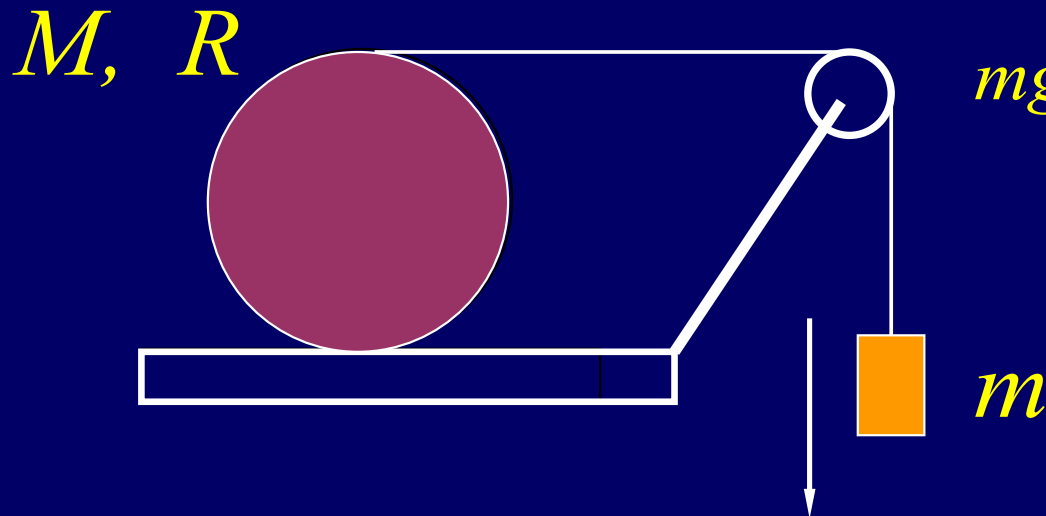
$$a = \frac{mg}{\frac{3}{8}M + m}$$



## Example

$$mgy = \frac{1}{2}mv^2 + \frac{1}{8}Mv^2 + \frac{1}{16}Mv^2$$

$$\omega R = v_c \quad v = 2v_c$$



Rolling without  
slipping

$$a = ?$$

$$mg \frac{dy}{dt} = mv \frac{dv}{dt} + \frac{1}{4}Mv \frac{dv}{dt} + \frac{1}{8}Mv \frac{dv}{dt}$$

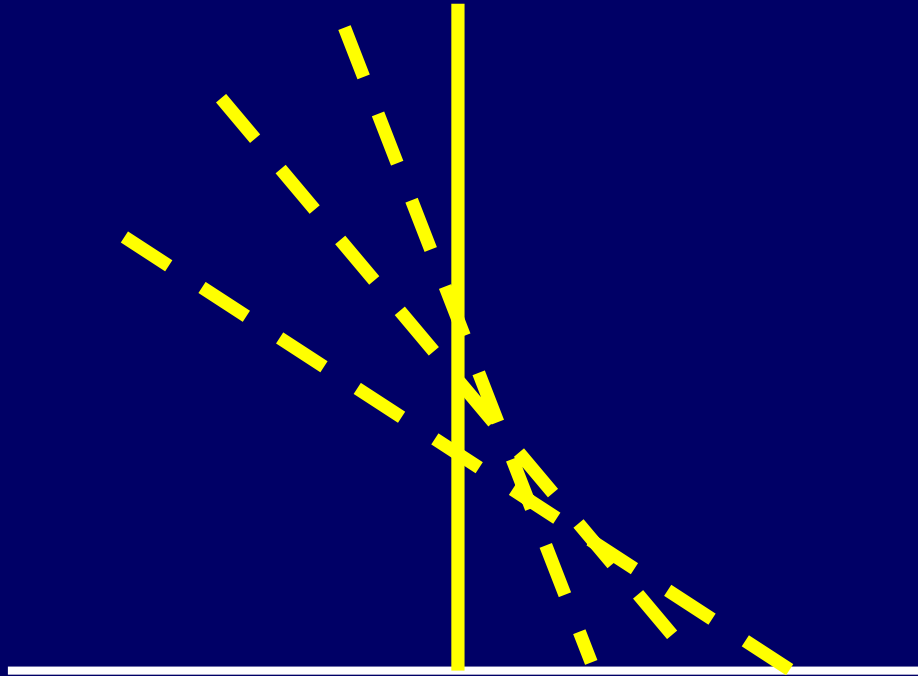
$$mg = ma + \frac{1}{4}Ma + \frac{1}{8}Ma$$

$$a = \frac{mg}{\frac{3}{8}M + m}$$

$$2\omega R = v \quad mgy = \frac{1}{2}mv^2 + \frac{3}{16}Mv^2$$

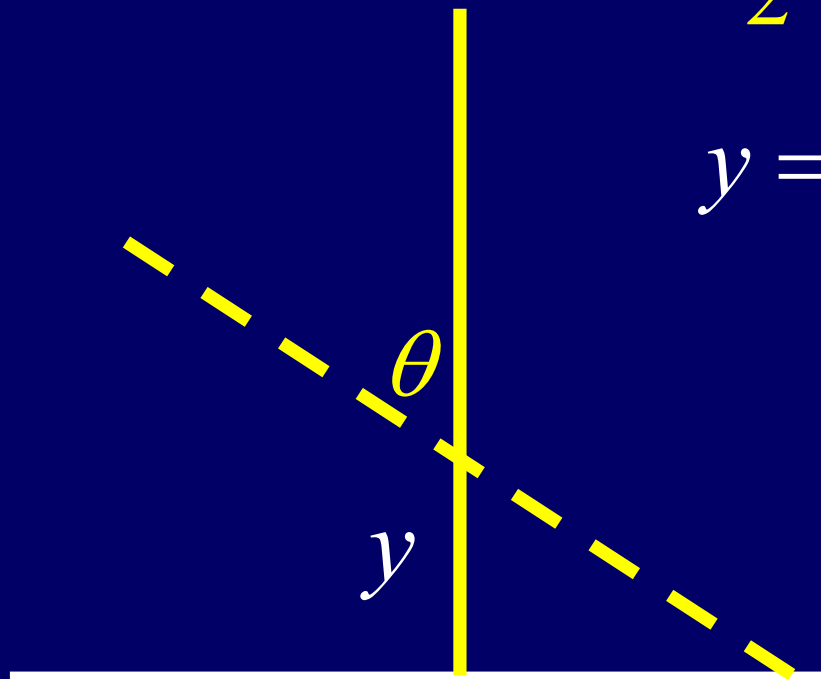
$$mg \frac{dy}{dt} = mv \frac{dv}{dt} + \frac{3}{8}Mv \frac{dv}{dt}$$

$m, L$



$$\omega = ?$$

$m, L$



$$\frac{mgL}{2}(1 - \cos \theta) = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2$$

$$y = \frac{L}{2} \cos \theta \quad \frac{dy}{dt} = -\frac{L}{2} \sin \theta \frac{d\theta}{dt}$$

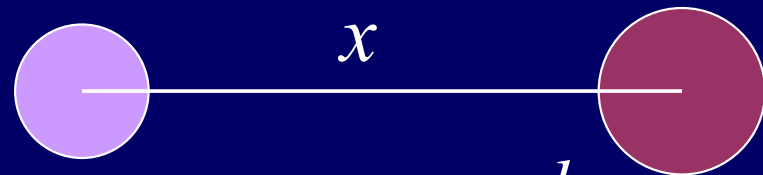
$$v_{cm} = -\frac{L}{2} \omega \sin \theta$$

$$\frac{mgL}{2}(1 - \cos \theta)$$

$$\omega = \sqrt{\frac{12g(1 - \cos \theta)}{L(3 \sin^2 \theta + 1)}} = \frac{1}{8}mL^2\omega^2 \sin^2 \theta + \frac{1}{24}mL^2\omega^2$$

$$\omega|_{\theta=0} = 0 \quad \omega|_{\theta=90^\circ} = \sqrt{\frac{3g}{L}}$$

Example: The potential energy function for the force between two atoms in a diatomic molecule can be expressed approximately as follows:



$$U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$$

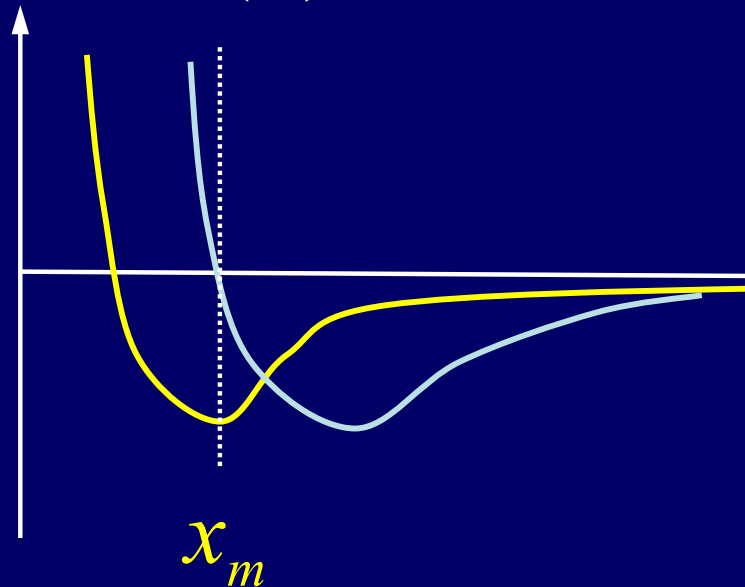
$$a = \text{const.} \quad b = \text{const.}$$

Find (1) the force between the atoms. (2) the equilibrium separation between the atoms. (3) the minimum energy necessary to break the molecule apart.

$$(3) U(x_m) = \frac{a}{x_m^{12}} - \frac{b}{x_m^6} = -\frac{b^2}{4a}$$

$$E_d = \frac{b^2}{4a} \quad U(\infty) = \frac{a}{x^{12}} - \frac{b}{x^6} = 0$$

$U(x)$   $F(x)$



$$(2) \frac{12a}{x_m^{13}} - \frac{6b}{x_m^7} = 0 \quad x_m = \left( \frac{2a}{b} \right)^{1/6}$$

CHAP.12

Exercises

P272:7, 9, 15